

Supplementary Material

Catalogue of Catalogues: A Web Platform for Earthquake Catalogue Management,
Merging, and Statistical Analysis

This supplement collects additional screen captures of the *Catalogue of Catalogues* (CofC) platform that support, but are not essential to, the argument of the main paper. All captures are from release v0.1.0 running in a browser against a local instance.

Data. Every capture uses the reduced-scale synthetic catalogue pair described in the main paper (a 5,200-event “GeoNet-like” catalogue and a 4,300-event “Agency B” catalogue, sharing a 50 % duplicate overlap), generated with a fixed seed by the script in the source repository. No real event data are shown anywhere in this supplement.

Statistical analysis in the browser

Event-level quality index

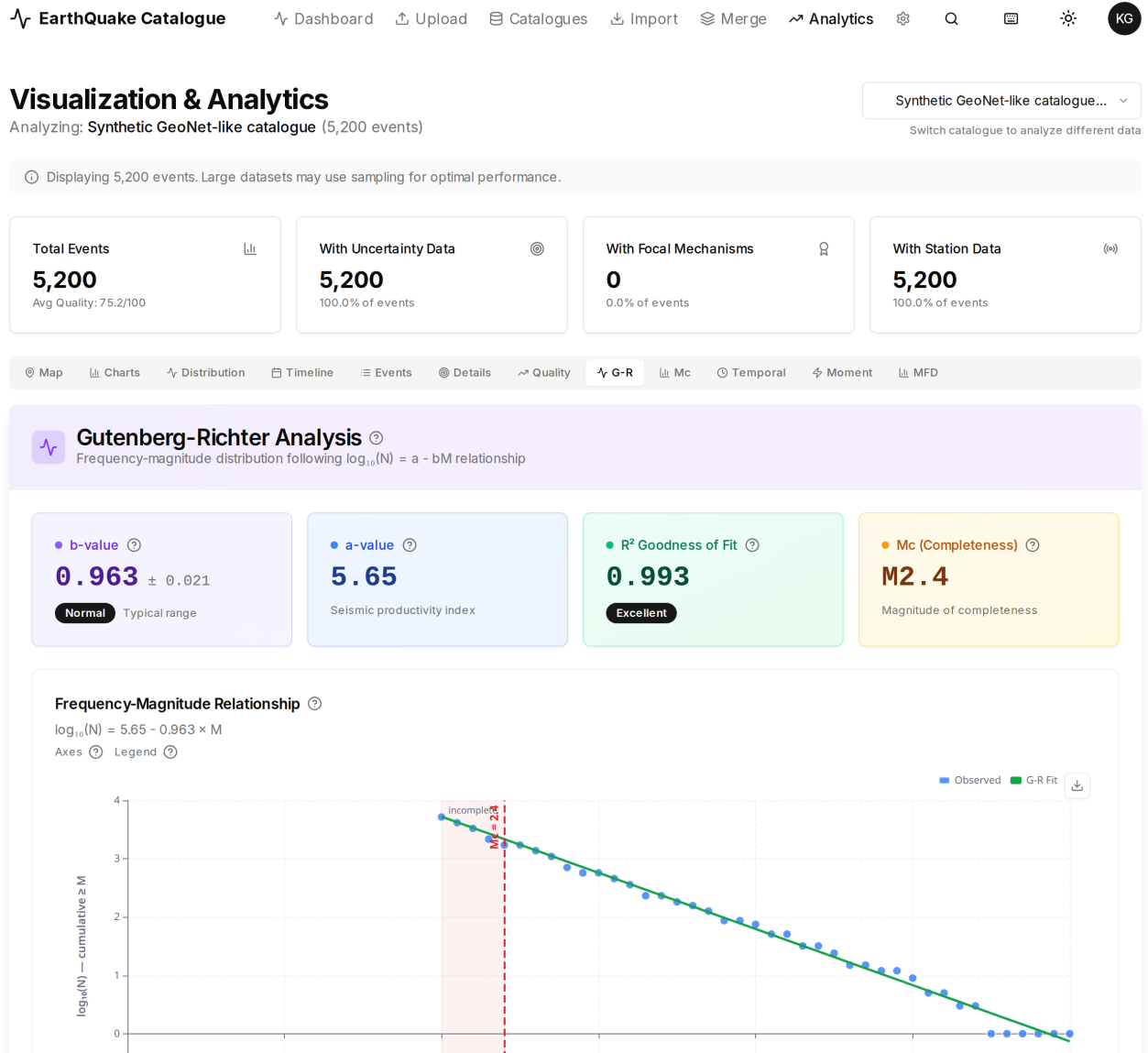


Figure S1: Gutenberg-Richter analysis of the synthetic GeoNet-like catalogue, computed client-side. The platform reports $\hat{b}$ with its counting-statistics uncertainty, the a -value, the goodness of fit, and the Maximum Curvature estimate of M_c ; the frequency-magnitude plot marks the incomplete range below M_c and overlays the fitted relation. The recovered $\hat{b} = 0.963 \pm 0.021$ is consistent with the $b = 1.0$ used to generate the synthetic catalogue, and is a check on the estimator rather than a measurement of New Zealand seismicity.

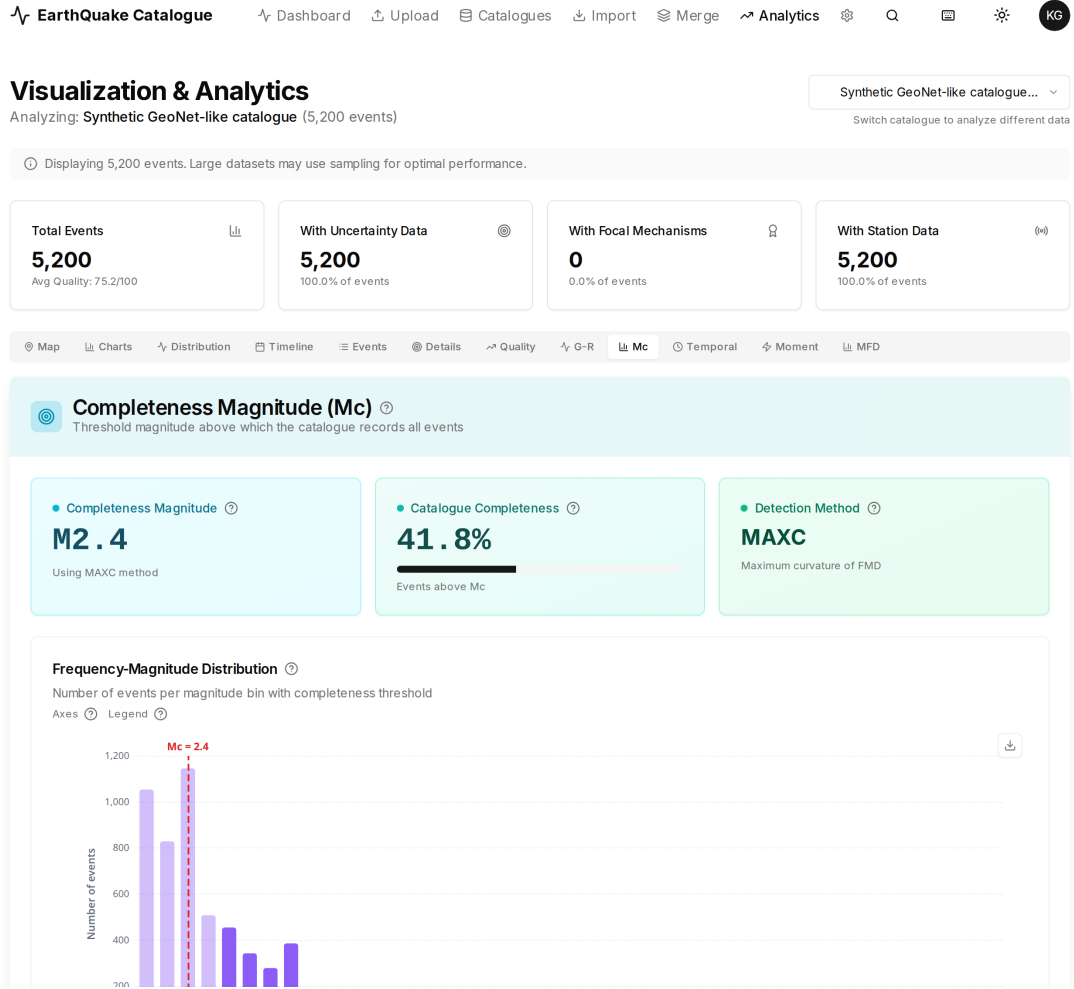


Figure S2: Magnitude-of-completeness panel for the same catalogue. M_c is estimated by Maximum Curvature with the +0.2 correction discussed in the main paper, and is exposed as an explicit, user-adjustable cut-off rather than applied silently, so that the completeness assumption behind any subsequent b -value is visible at the point of use.

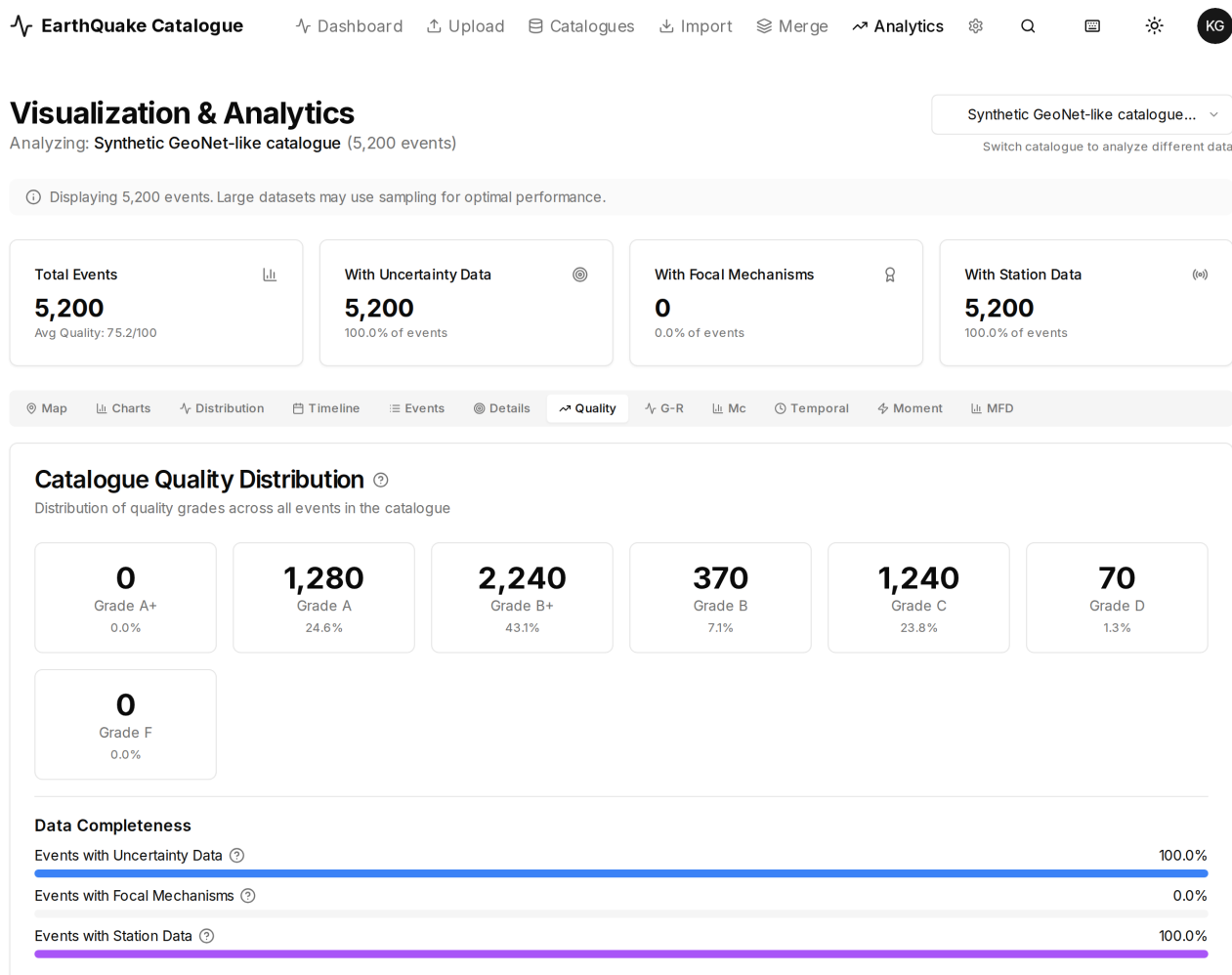


Figure S3: Catalogue-level summary of the per-event quality index (equation 1 of the main paper) for the synthetic GeoNet-like catalogue. Every event is scored on import and mapped to an A⁺–F grade; the panel reports the grade distribution and the mean score, together with the field-completeness fractions that feed the individual sub-scores. The absence of focal mechanisms in this synthetic catalogue is reflected in the completeness bars, and depresses the achievable score, which is why no event reaches grade A⁺.